Getting started with the `abmneo()` function

Sasha D. Hafner

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abmneo

The abmneo package, with a single `abmneo()` function, is an implementation of the ABM model for anaerobic biodegradation of organic wastes (Dalby et al., 2021). The ABM model is already implemented in an earlier ABM package, but that package codebase is relatively large and complex. The new package was developed as a simpler but more flexible alternative. Most of that simplicity is in the code; the user experience is roughly similar if default parameter sets are used. For user-defined substrates, microbial groups, or metabolic pathways, the newer package provides a lot flexibility. This vignette demonstrates the use of the `abmneo()` model function.

What's different?

The main differences between abmneo and ABM are:

1. package complexity and hard-coding,
2. handling of time-variable inputs, and
3. input flexibility for substrates, microbial groups, and stoichiometry.

Not all of these are discussed in this document.

In abmneo, time-variable inputs (2) are treated using steps instead of the interpolation approach used in ABM. This approach ostensibly reduces the accuracy of the model as a tradeoff for simpler code, more flexibility in which inputs can vary, and faster evaluation. But in practice the effects seem negligible beyond a visual effect in plots.

Hard-coded behavior and special cases in ABM includes the microbial community, with fixed symbols for methanogen and sulfate reducer names and designated code for each. In contrast, the microbial community in abmneo is defined completely by input parameters, and stoichiometry can be specified. Internally, shorter vectorized or matrix operations do the work for all groups without special cases. Substrates and inhibition are two other areas where these types of major changes were made.

C++ code and the Rcpp package were incorporated into ABM for speed, but the cost is code complication and debugging difficulty, as well as increased requirements for installation. So it was dropped from abmneo, with any speed penalty offset by other simplifications. In the end abmneo is not obviously slower than ABM; it is even slightly faster in some cases. Other dependencies were also eliminated in abmneo.

```
devtools::load_all()
```

```
## i Loading abmneo
## Loading required package: deSolve
```

Input arguments and parameters

Here are the `abmneo()` arguments:

```
abmneo <- function(
  storage,
  days = 365,
  delta_t = 1,
  times = NULL,
  inf_pars = NULL,
  grp_pars = NULL,
  sub_pars = NULL,
  chem_pars = list(COD_conv = c(CH4 = 5.32), gases = c('CH4', 'CO2', 'H2S')),
  ctrl_pars = list(
    approx_method = 'early',
    fill_method = 'interp',
    par_key = '\\.',
    arr_max_temp_K = 313
  ),
  var_pars = list(var = NULL),
  add_pars = NULL,
  pars = NULL,
  startup = 0,
  starting = NULL,
  warn = TRUE
)
```

We'll come back to the `storage` argument later. The `days` (total simulation time), `delta_t` (time step *in output*), and `times` (specific times to be included in output) are similar in ABM and straightforward. Let's work on the `*_pars` arguments. These are all input parameters. In this first release, abmneo does not include any default parameter sets. Instead, this example presents a set similar to the ABM set `grp_pars_pig2.0` in ABM.

`inf_pars` has inflow properties. VFA is a hard-coded intermediate, and we need to set fresh and initial concentrations (g/kg) along with pH and density. The presence of any other solutes, with any names, is set in the `comps` element, with concentrations in g/kg given in `comp_fresh` and `comp_init` elements.

```
inf_ref <- list(VFA_fresh = 2, VFA_init = 2, comps = c('SO4'), comp_fresh = c(SO4 = 0.2), comp_init = c
```

`grp_pars` defines the microbial community. The argument should have these elements:

- `grps`: group names
- `yield`: biomass yields on COD basis
- `xa_fresh`: active biomass concentrations in fresh influent
- `xa_init`: initial concentrations
- `d_max`: maximum death/decay rate (1/d)
- `qhat_opt`: maximum substrate untilization rate (g COD / g COD - d)
- `T_opt`: temperature of maximum activity (deg. C)
- `T_min`: minimum temperature with any activity (deg. C)
- `T_max`: maximum temperature with any activity (deg. C)
- `ksmat`: matrix of Monod parameters
- `mstoich`: matrix of stoichiometry

Definition of the microbial community is completely flexible; any number of groups with any names an practically any metabolism. Here we'll include those groups used in the latest ABM parameter set. Let's start with the stoichiometry matrix.

```
mstoich <- matrix(
  c(
    -1, -1, -1, -1, -1, -1, -1,
```

```
    0,  0,  0,  0,  0,  0, -0.50156,
    1,  1,  1,  1,  1,  1,  0
  ),
  nrow = 3,
  byrow = TRUE,
  dimnames = list(
    c('VFA', 'SO4', 'CH4'),
    c('m0', 'm1', 'm2', 'm3', 'm4', 'm5', 'sr1')
  )
)
```

It is easier to understand by looking at the matrix with row and column names:

```
mstoich
```

```
##     m0 m1 m2 m3 m4 m5      sr1
## VFA -1 -1 -1 -1 -1 -1 -1.00000
## SO4  0  0  0  0  0  0 -0.50156
## CH4  1  1  1  1  1  1  0.00000
```

Columns are groups, rows are substrates or products. The values of 1 are an expression of chemical oxygen demand (COD) conservation–they show that each methanogen group produces 1 g of methane per g of VFA consumed (both on a COD basis). Compounds without a COD are expressed as mass N (if present), S (if present), or otherwise total compound mass. Here then `'SO4-2'` is as mass of S. So where did the `0.50` come from? The micstoich package was used to sort out the stoichiometry of sulfate reduction.

```
library(micstoich)
rxn <- micstoich('CH3COOH', acceptor = 'SO4-2', product = 'H2S')
rxn
molmass('SO4-2', 'S') * rxn['SO4-2'] / (calcCOD('CH3COOH', per = 'mol') * rxn['CH3COOH'])
```

This gives:

```
> library(micstoich)
> rxn <- micstoich('CH3COOH', acceptor = 'SO4-2', product = 'H2S')
> rxn
CH3COOH   SO4-2      H+     H2S     HS-     H2O     CO2
-0.1250 -0.1250 -0.1875  0.0625  0.0625  0.2500  0.2500
> molmass('SO4-2', 'S') * rxn['SO4-2'] / (calcCOD('CH3COOH', per = 'mol') * rxn['CH3COOH'])
    SO4-2
0.5015625
```

Note that these coefficients do not account for microbial biomass–this information is entered as a microbial yield, and the stoichiometry matrix is updated and expanded internally.

We need one more matrix: `ksmat`, which has the Monod half-velocity parameters. Here the values are from `mic.pars2.0` in the ABM repo

```
ksmat <- matrix(
  c(1.15, 1.15, 1.15, 1.15, 1.15, 1.15, 0.46,
      NA,   NA,   NA,   NA,   NA,   NA, 0.00694),
  nrow = 2,
  byrow = TRUE,
  dimnames = list(
    c('VFA', 'SO4'),
    c('m0', 'm1', 'm2', 'm3', 'm4', 'm5', 'sr1')
  )
)
```

```
ksmat
```

```
##       m0   m1   m2   m3   m4   m5      sr1
## VFA 1.15 1.15 1.15 1.15 1.15 1.15 0.46000
## SO4   NA   NA   NA   NA   NA   NA 0.00694
```

Note that `NA` values for sulfate and the methanogens.

Now let's fill in a `grp_pars` list. Unlike the ABM package, there is no `all` keyword in abmneo, but the `default` keyword behaves the same way where no group is expicitly specified.

```
grp_ref <- list(
  grps = c('m0', 'm1', 'm2','m3','m4','m5','sr1'),
  yield = c(default = 0.05, sr1 = 0.065),
  xa_fresh = c(default = 0.06),
  xa_init = c(default = 0.1),
  d_max = c(default = 0.02),
  qhat_opt =  c(m0 = 0.463, m1 = 0.746, m2 = 0.901, m3 = 2.80, m4 = 4.66, m5 = 7.46, sr1 = 8.34),
  T_opt = c(m0 = 18, m1 = 18,  m2 = 28,  m3 = 36, m4 = 44, m5 = 55, sr1 = 44),
  T_min = c(m0 =  0, m1 =  9.7, m2 = 9.7, m3 = 15, m4 = 26, m5 = 30, sr1 =  0),
  T_max = c(m0 = 25, m1 = 25,  m2 = 38,  m3 = 45, m4 = 51, m5 = 60, sr1 = 51),
  ksmat = ksmat,
  mstoich = mstoich
)
```

Substrates are completely defined in the `sub_pars` input.

```
sub_ref <- list(
  subs = c('starch', 'Cfat', 'CPs', 'CPf', 'RFd'),
  sub_enrich = c('starch', 'Cfat', 'CPs', 'RFd'),
  xd = 'xd',
  arrA = c(starch = 1.005e13, Cfat = 2.011e13, CPs = 4.152e12, CPf = 4.152e12, RFd = 2.011e12, xd = 2.0
  arrE = c(default = 8.16e4),
  sub_fresh = c(starch = 5.25, Cfat = 27.6, CPs = 11.6, CPf = 9.50, RFd = 25.4, xd = 0),
  sub_init  = c(starch = 5.25, Cfat = 27.6, CPs = 11.6, CPf = 9.50, RFd = 25.4, xd = 0)
)
```

So now we've created lists for `inf_pars`, `grp_pars`, and `sub_pars`. These are the only required parameter groups.

The `chem_pars` argument has COD conversion ratios in g COD per g output or input units. These are needed for the COD balance and expressing output in useful units. The default element is for methane, as g COD per g methane C. Of course VFA and substrates are all in COD units, so there is no conversion. The `gases` element names the components that should be treated as gases, they do not accumulate in the slurry. Cumulative emission and emission rate is calculated for each gas. All other components are solutes, which accumulate until the storage is emptied.

The `var_pars` argument is used for time-variable inputs. We won't use it in the first example. And `add_pars` is for adding or modifying parameters set through other arguments. We won't use that either.

But we do need to sort out the `storage` argument. There are two basic approaches that can be used: "regular", with a fixed rate of influent addition and a fixed emptying interval; and "series", where influent addition and storage emptying is completely flexible, and defined by a slurry level over time.

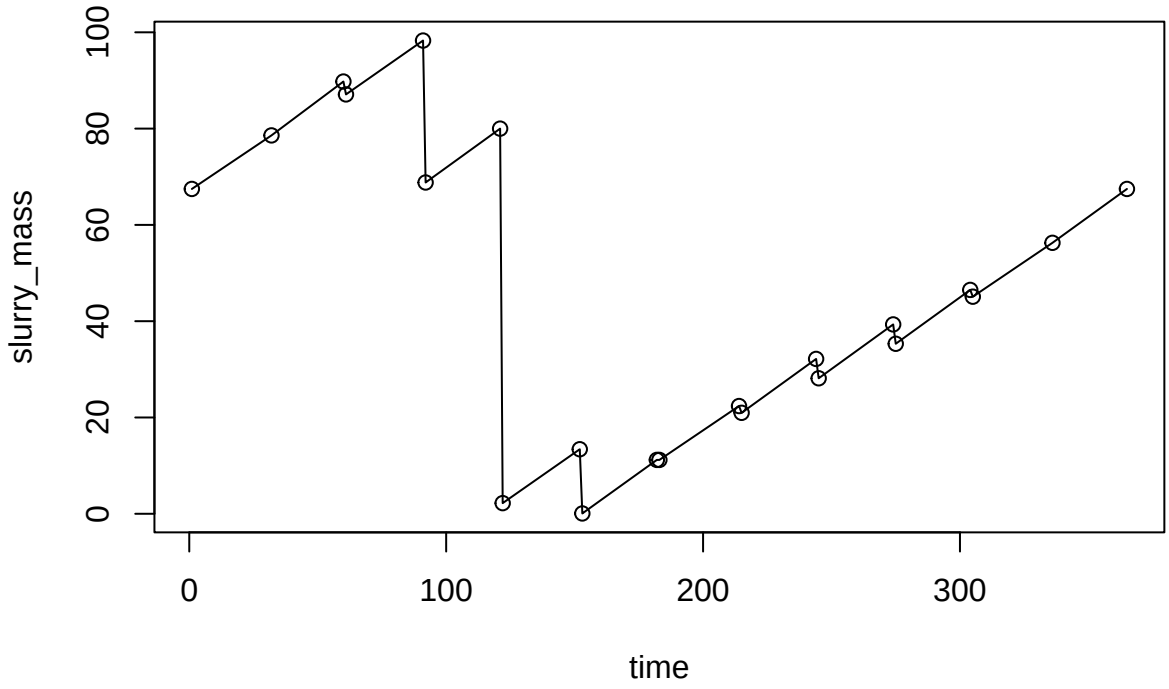
Regular option

For the regular option, `abmneo()` needs a list with these elements:

```
stor_reg_ref <- list(
  type = 'regular',
  slurry_prod_rate = 5,
  storage_depth = 2,
  area = 0.65,
  temp_C = 20,
  resid_enrich = 0.2,
  slurry_mass = 7,
  resid_depth = 0.01,
  empty_int = 28,
  wash_water = 0,
  wash_int = 90,
  rest_d = 1
)
```

And for a series, the main element is a data frame with the measurements.

```
mass_series <- read.csv('level_pig_od_ref.csv')
plot(slurry_mass ~ time, data = mass_series, type = 'o')
```




```
stor_ser_ref <- list(
  type = 'series',
  dat = mass_series,
  storage_depth = 2,
  area = 0.7,
  temp_C = 18,
  resid_enrich = 0.2
```

```
)
```

These are both for pig production systems.

Finally, let's run the model.

```
devtools::load_all()
```

```
## i Loading abmneo
```

```
out_reg <- abmneo(
  storage = stor_reg_ref,
  365,
  inf_pars = inf_ref,
  grp_pars = grp_ref,
  sub_pars = sub_ref
)
```

```
## Warning in pack_pars(storage = storage, inf_pars = inf_pars, grp_pars =
## grp_pars, : Expanding mstoich matrix with new rows to include missing comps or
## gases.
```

Output is a data frame that contains gas emission rates, cumulative emission, and masses and concentrations of microbial biomass, substrates, VFA, and any other solutes. The column names, and number of columns, depends on the inputs; each microbial group and each substrate has two associated columns.

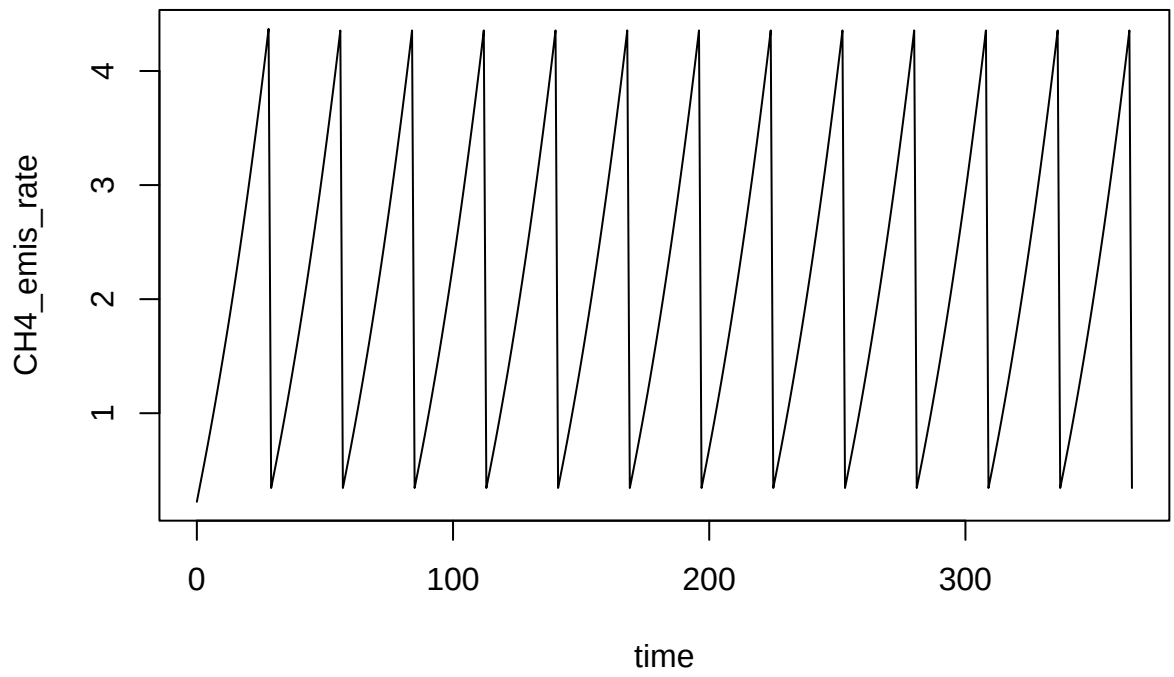
```
names(out_reg)
```

```
##  [1] "time"              "m0"                "m1"
##  [4] "m2"                "m3"                "m4"
##  [7] "m5"                "sr1"               "starch"
## [10] "Cfat"              "CPs"               "CPf"
## [13] "RFd"               "xd"                "VFA"
## [16] "SO4"               "slurry_mass"       "CH4"
## [19] "CO2"               "H2S"               "slurry_load"
## [22] "COD_load"          "CH4_emis_rate"     "temp_C"
## [25] "pH"                "m0_eff"            "m1_eff"
## [28] "m2_eff"            "m3_eff"            "m4_eff"
## [31] "m5_eff"            "sr1_eff"           "starch_eff"
## [34] "Cfat_eff"          "CPs_eff"           "CPf_eff"
## [37] "RFd_eff"           "xd_eff"            "VFA_eff"
## [40] "SO4_eff"           "slurry_mass_eff"   "slurry_depth"
## [43] "m0_conc"           "m1_conc"           "m2_conc"
## [46] "m3_conc"           "m4_conc"           "m5_conc"
## [49] "sr1_conc"          "starch_conc"       "Cfat_conc"
## [52] "CPs_conc"          "CPf_conc"          "RFd_conc"
## [55] "xd_conc"           "VFA_conc"          "SO4_conc"
## [58] "m0_eff_conc"       "m1_eff_conc"       "m2_eff_conc"
## [61] "m3_eff_conc"       "m4_eff_conc"       "m5_eff_conc"
## [64] "sr1_eff_conc"      "starch_eff_conc"   "Cfat_eff_conc"
## [67] "CPs_eff_conc"      "CPf_eff_conc"      "RFd_eff_conc"
## [70] "xd_eff_conc"       "VFA_eff_conc"      "SO4_eff_conc"
## [73] "CH4_emis_rate_ave"
```

The plots below show some results.

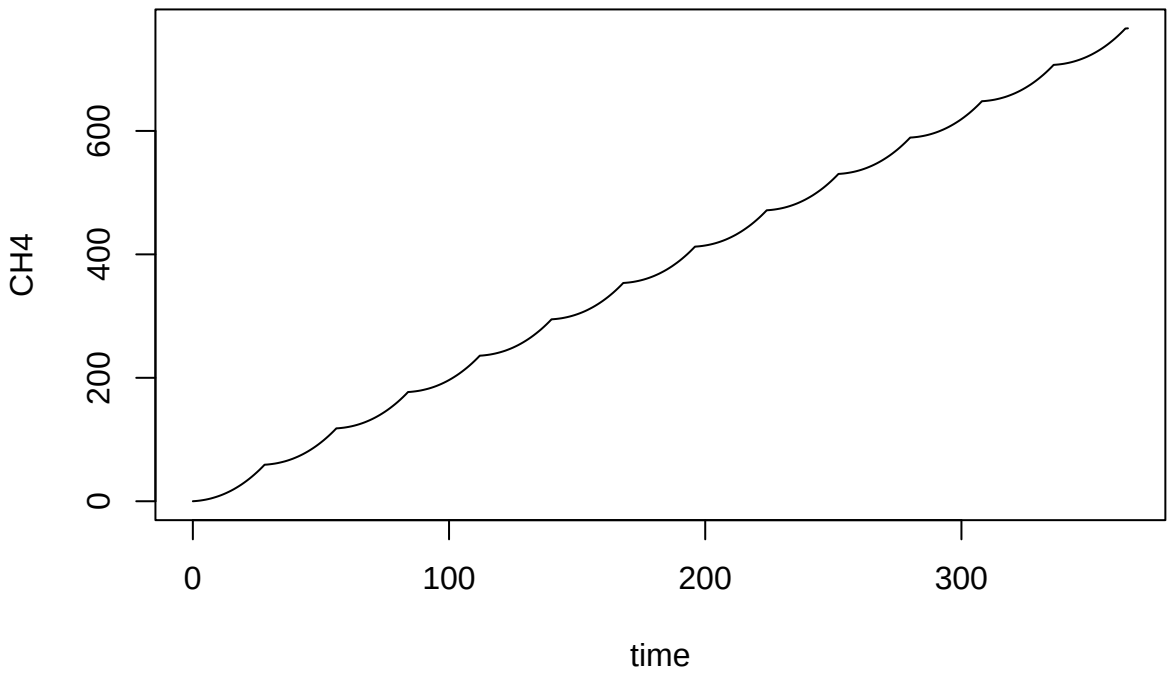
Methane emission rate, in g/d.

```
plot(CH4_emis_rate ~ time, data = out_reg, type = 'l')
```



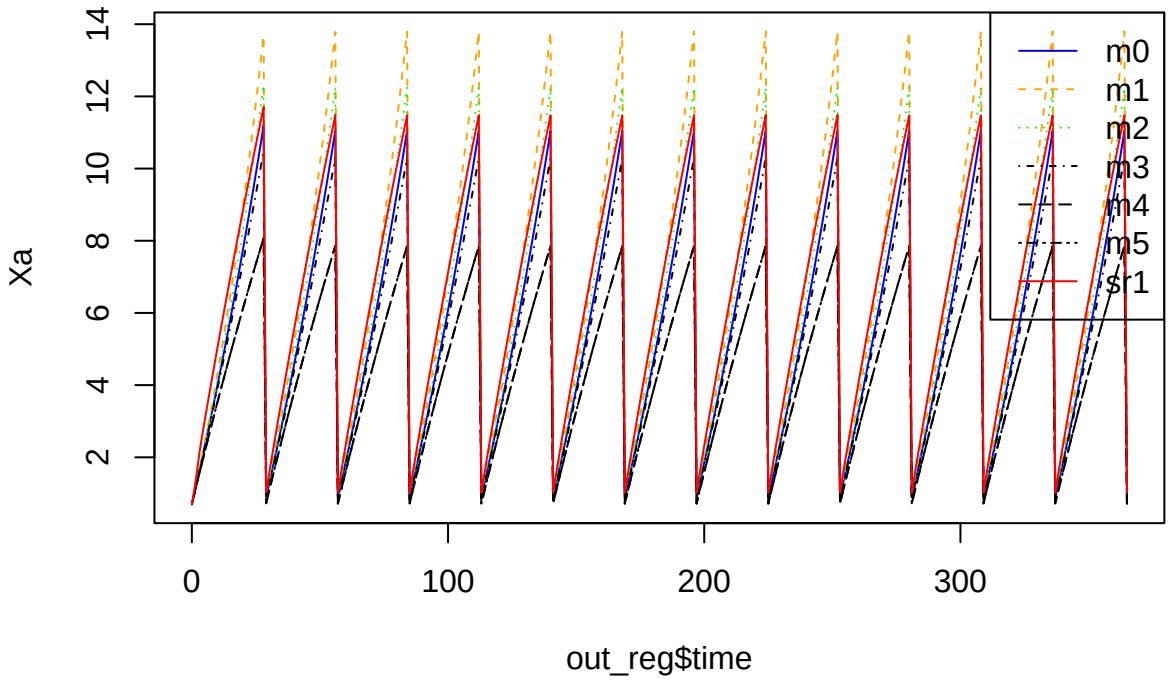
And cumulative emission in g.

```
plot(CH4 ~ time, data = out_reg, type = 'l')
```



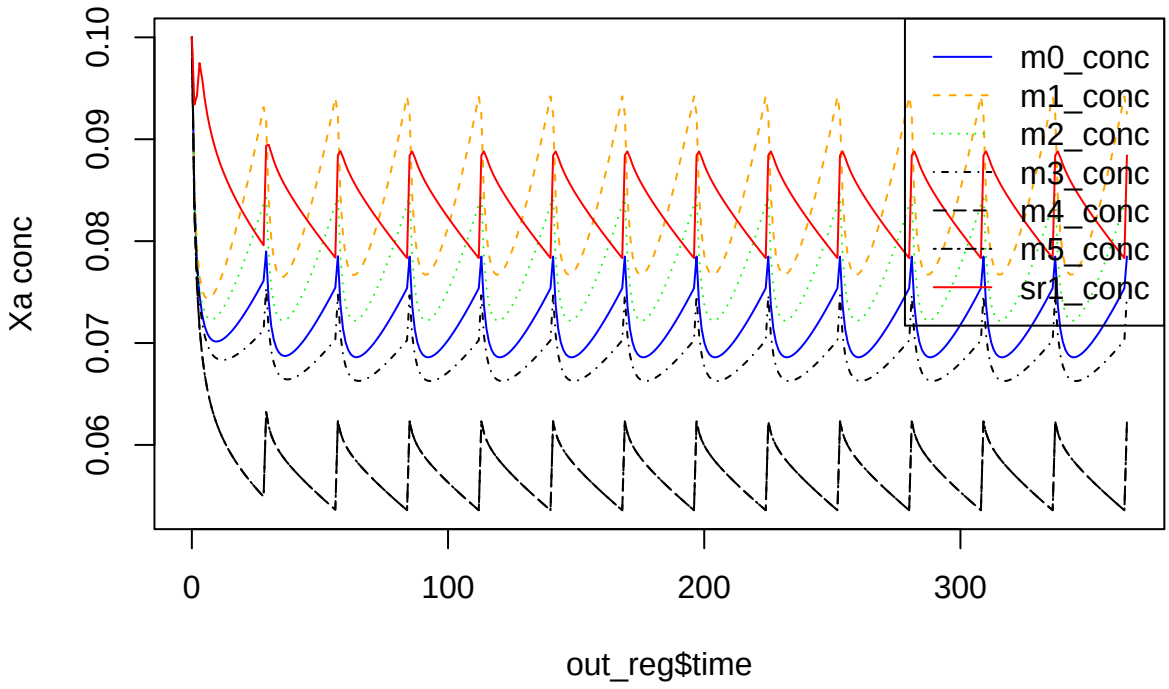
Microbial biomass (g).

```
matplot(
  out_reg$time,
  out_reg[, nn <- grp_ref$grps],
  col = cc <- c('blue', 'orange', 'green', 'black', 'black', 'black', 'red'),
  lty = 1:length(nn), type = 'l',
  ylab = 'Xa'
)
legend('topright', legend = nn, col = cc, lty = 1:length(nn))
```



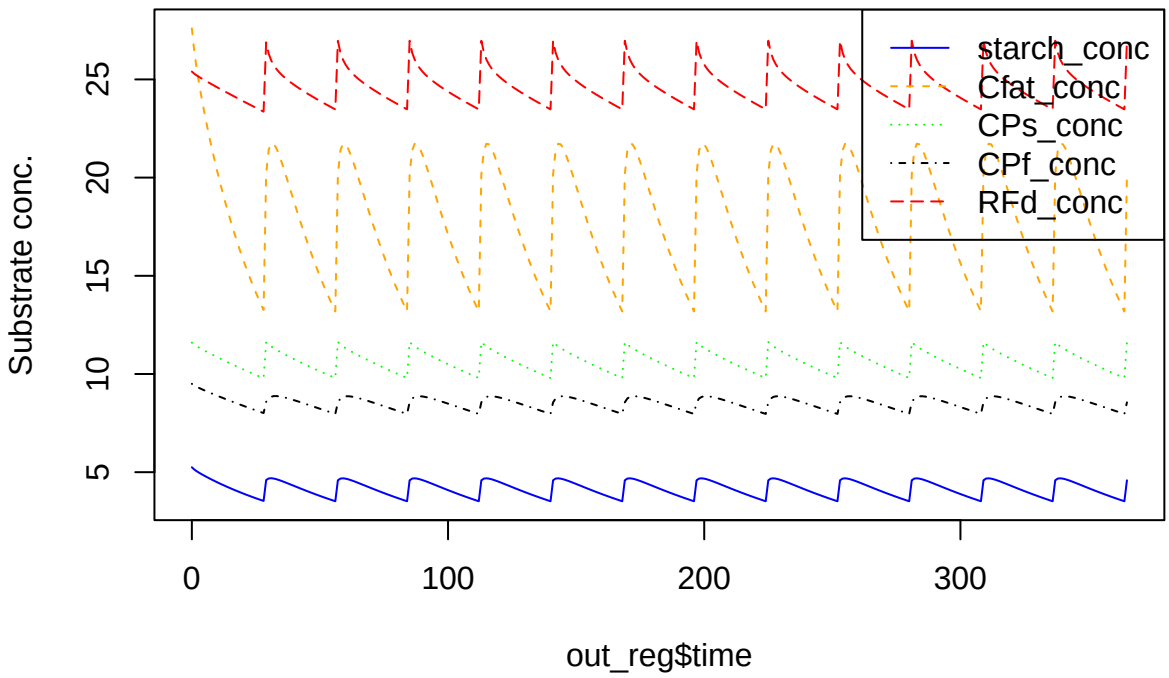
And microbial biomass concentrations (g/kg).

```
matplot(
  out_reg$time,
  out_reg[, nn <- paste0(grp_ref$grps, '_conc')],
  col = cc <- c('blue', 'orange', 'green', 'black', 'black', 'black', 'red'),
  lty = 1:length(nn),
  type = 'l',
  ylab = 'Xa conc'
)
legend('topright', legend = nn, col = cc, lty = 1:length(nn))
```



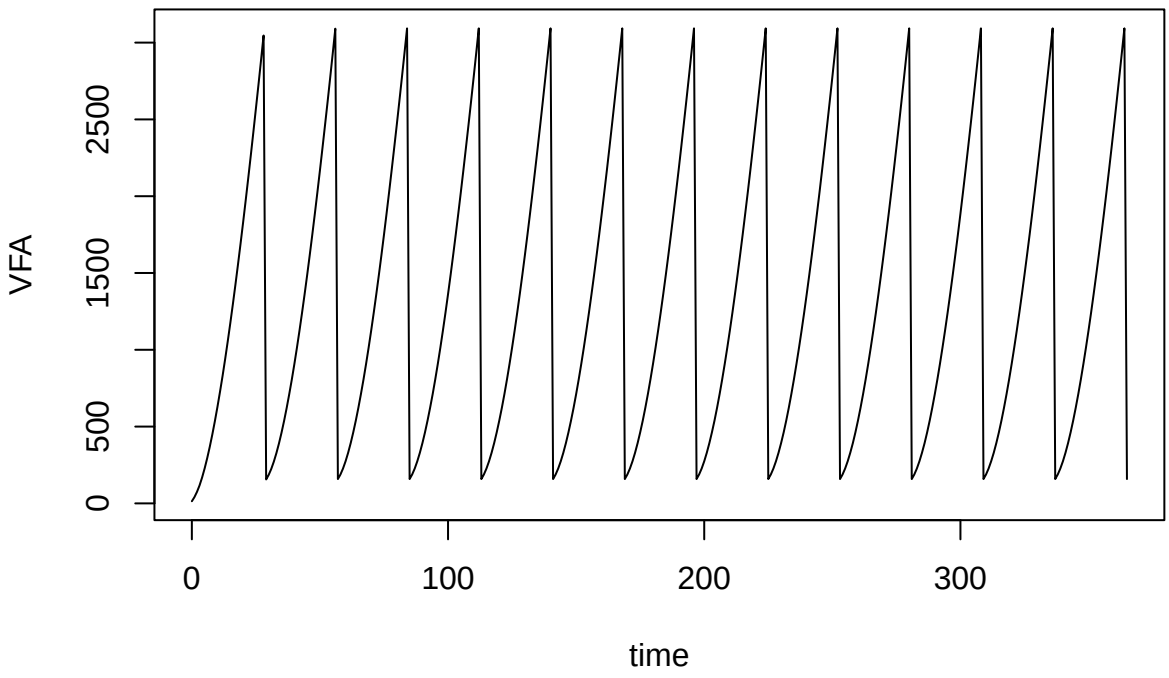
Substrate concentrations.

```
matplot(
  out_reg$time,
  out_reg[, nn <- paste0(sub_ref$subs, '_conc')],
  col = cc <- c('blue', 'orange', 'green', 'black', 'red'),
  lty = 1:length(nn),
  type = 'l',
  ylab = 'Substrate conc.'
)
legend('topright', legend = nn, col = cc, lty = 1:length(nn))
```

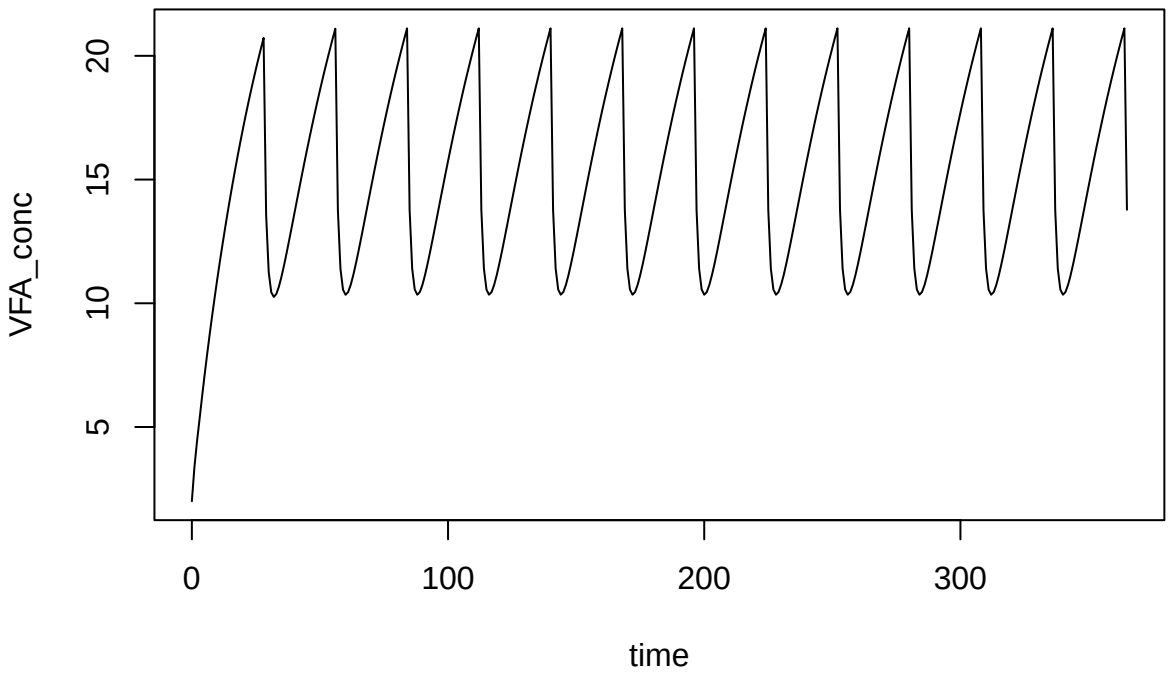


And VFA mass and concentration.

```
plot(VFA ~ time, data = out_reg, type = 'l')
```

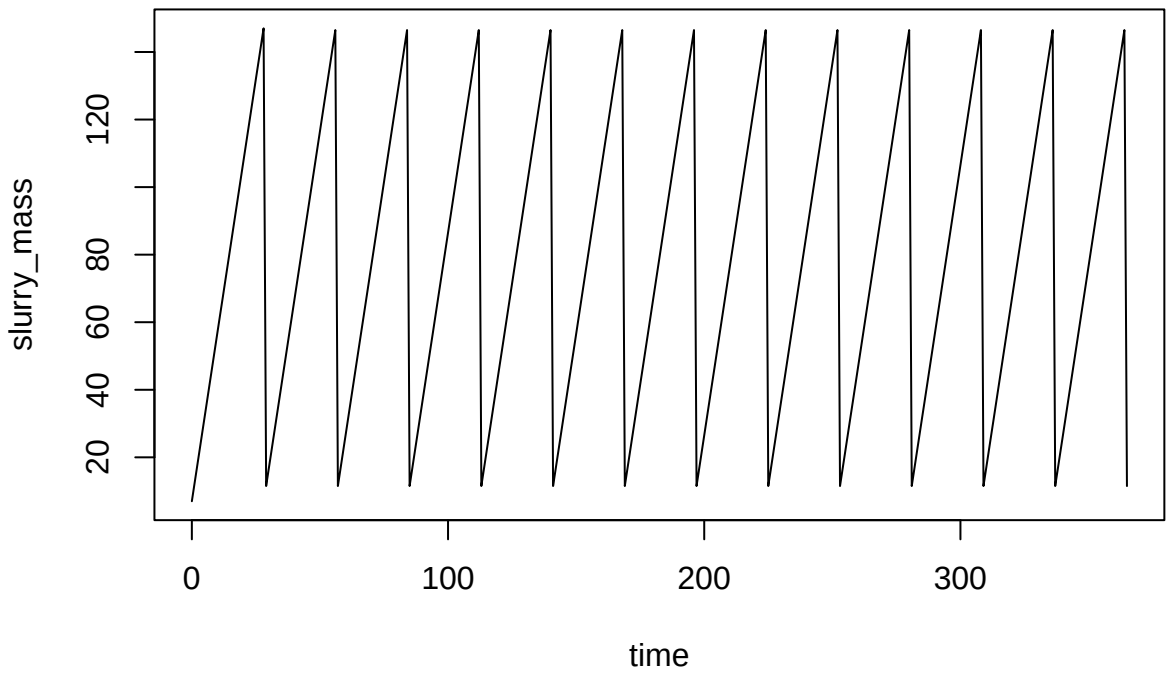



```
plot(VFA_conc ~ time, data = out_reg, type = 'l')
```



Of course most of the temporal patterns are driven by slurry mass, which is also returned.

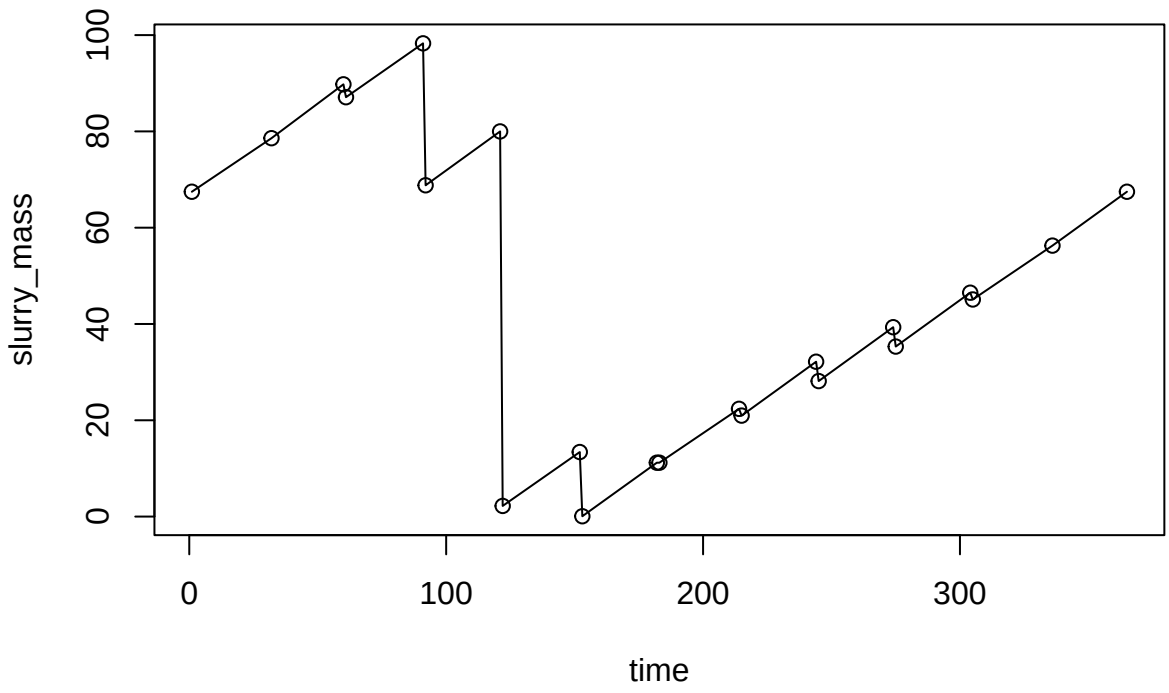
```
plot(slurry_mass ~ time, data = out_reg, type = 'l')
```



Time series option

For time series, we need a data frame with time and slurry mass columns, in that order. Here are some values.

```
mass_series <- read.csv('level_pig_od_ref.csv')
plot(slurry_mass ~ time, data = mass_series, type = 'o')
```



They are combined with other parameters like this.

```
stor_ser_ref <- list(
  type = 'series',
  dat = mass_series,
  storage_depth = 2,
  area = 0.03,
  temp_C = 15,
  resid_enrich = 0.9
)
```

```
devtools::load_all()
```

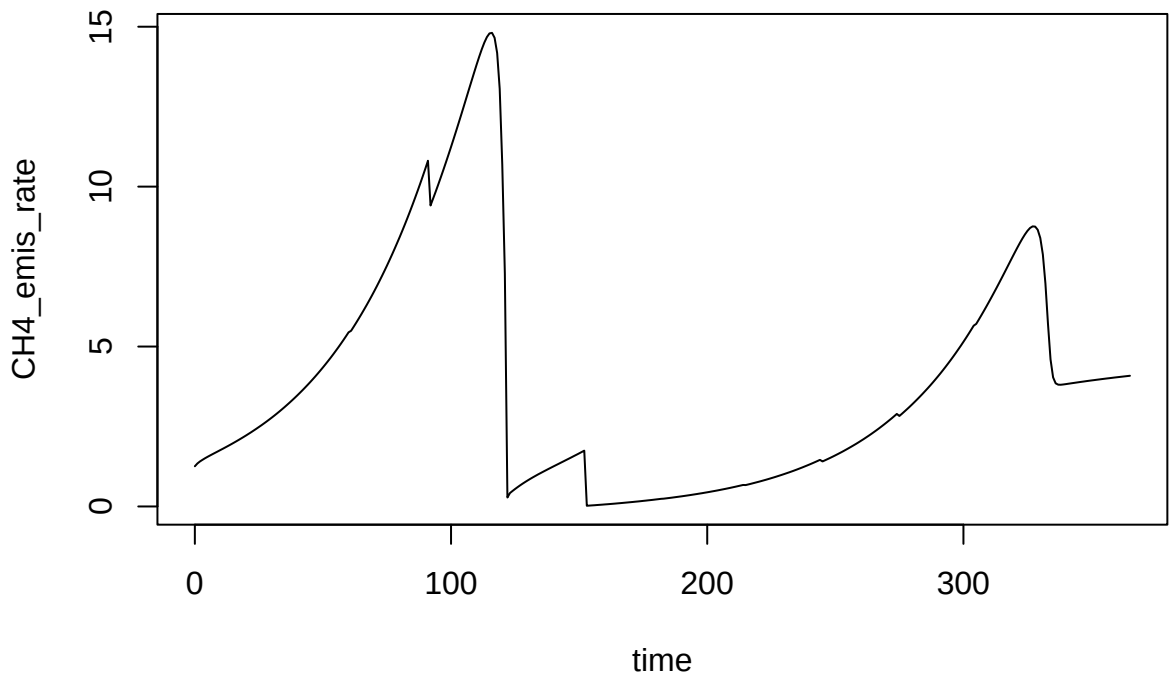
```
## i Loading abmneo
```

```
out_ser <- abmneo(
  storage = stor_ser_ref,
  365,
  inf_pars = inf_ref,
  grp_pars = grp_ref,
  sub_pars = sub_ref
)
```

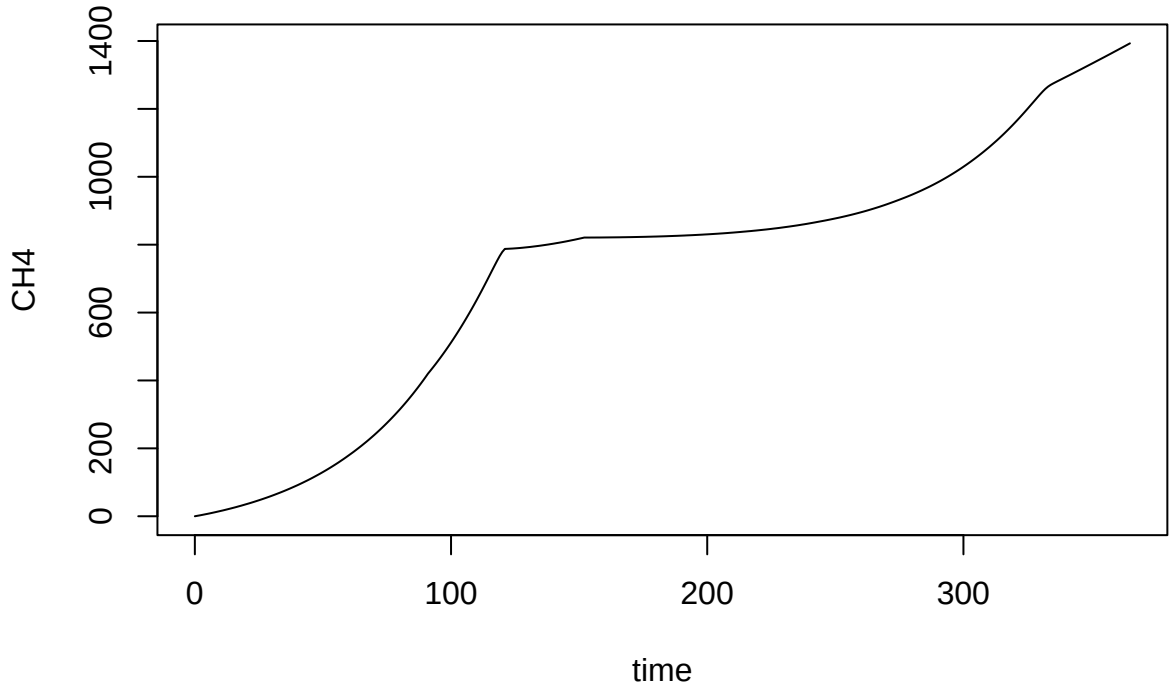
```
## Warning in pack_pars(storage = storage, inf_pars = inf_pars, grp_pars =
## grp_pars, : Expanding mstoich matrix with new rows to include missing comps or
## gases.
```

Here are the same plots that were shown above for regular inputs.

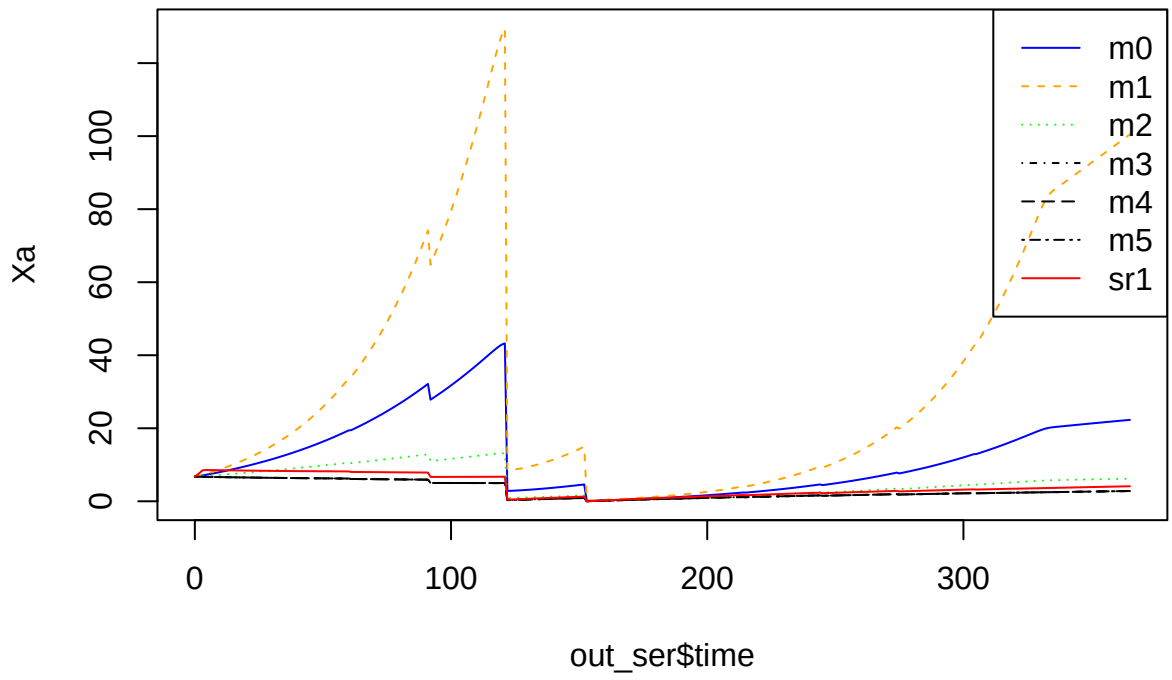
```
plot(CH4_emis_rate ~ time, data = out_ser, type = 'l')
```



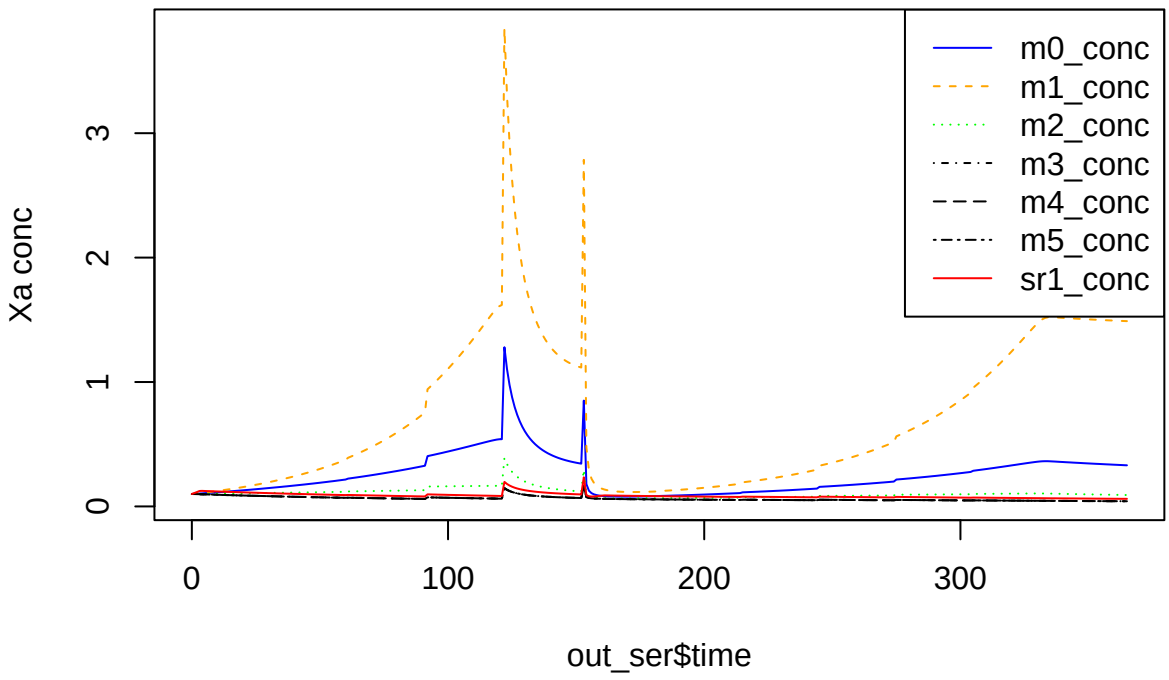

```
plot(CH4 ~ time, data = out_ser, type = 'l')
```



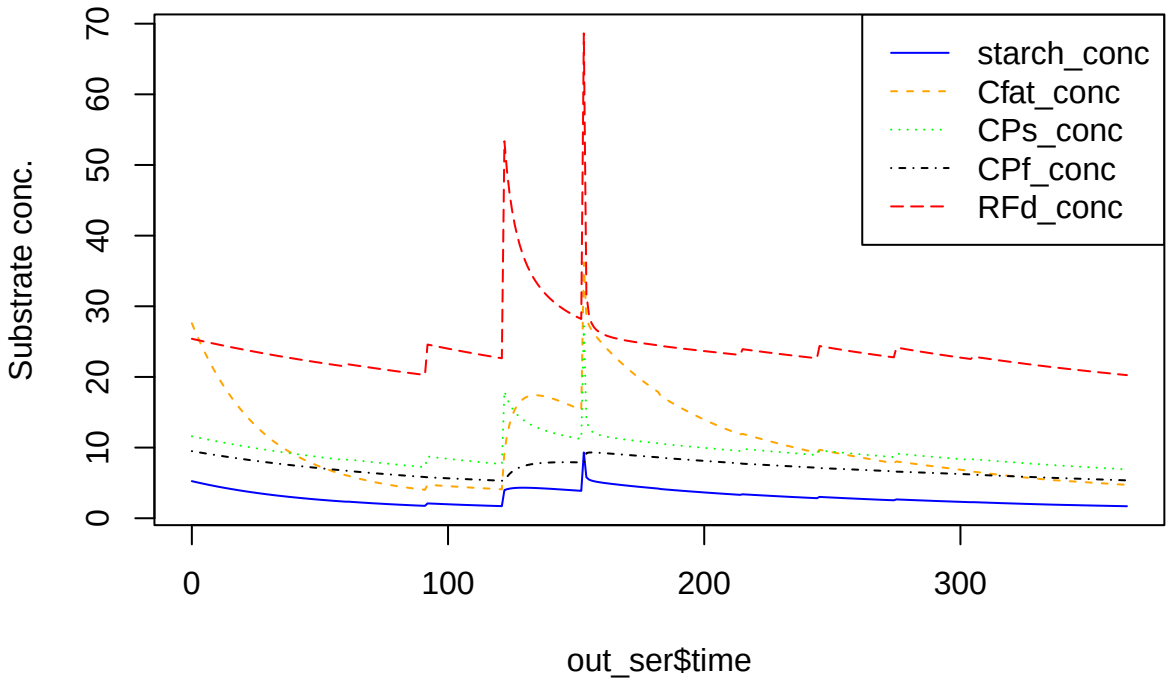

```
matplot(
  out_ser$time,
  out_ser[, nn <- grp_ref$grps],
  col = cc <- c('blue', 'orange', 'green', 'black', 'black', 'black', 'red'),
  lty = 1:length(nn), type = 'l',
  ylab = 'Xa'
)
legend('topright', legend = nn, col = cc, lty = 1:length(nn))
```



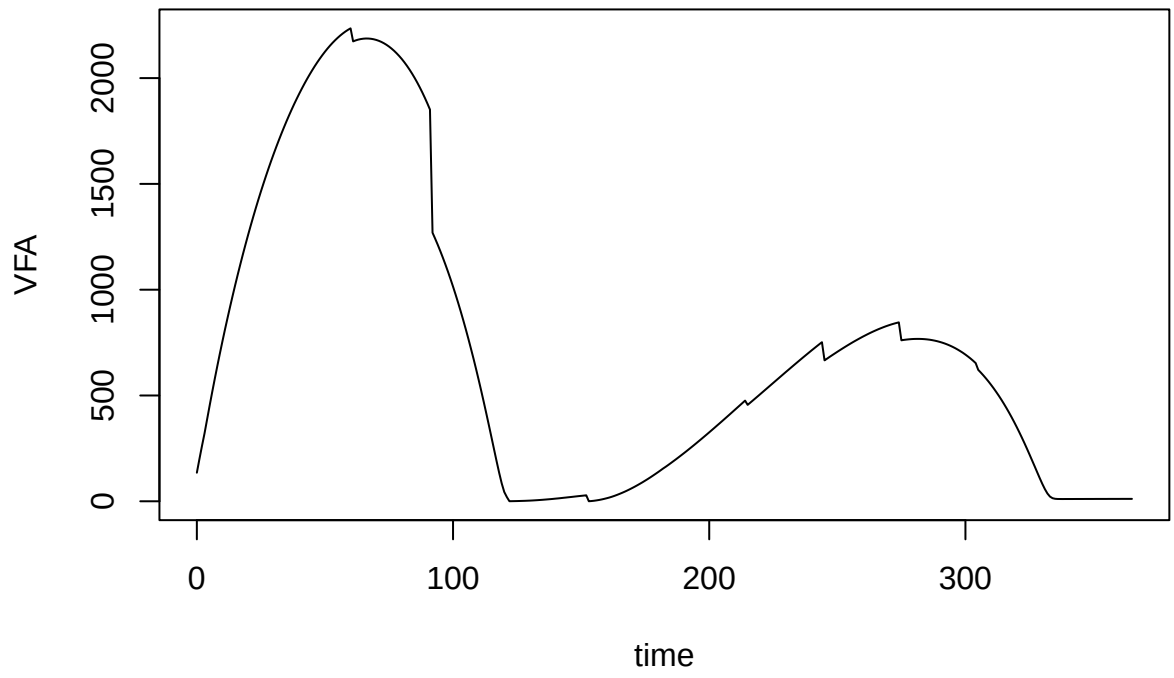

```
matplot(
  out_ser$time,
  out_ser[, nn <- paste0(grp_ref$grps, '_conc')],
  col = cc <- c('blue', 'orange', 'green', 'black', 'black', 'black', 'red'),
  lty = 1:length(nn),
  type = 'l',
  ylab = 'Xa conc'
)
legend('topright', legend = nn, col = cc, lty = 1:length(nn))
```



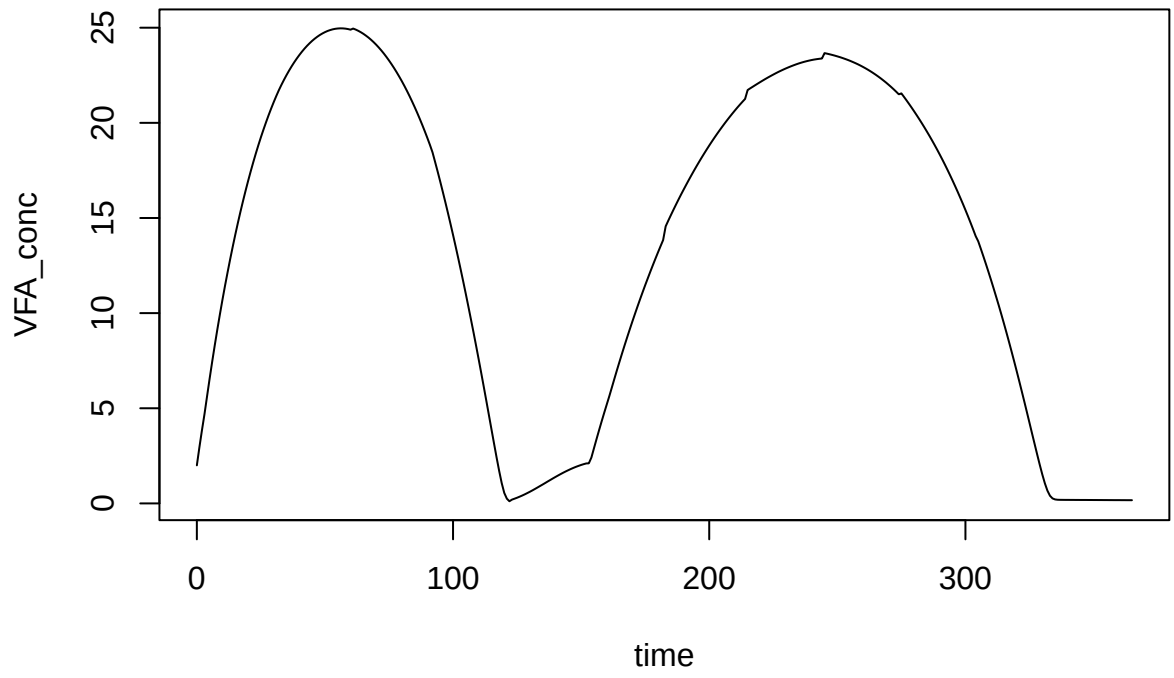

```
matplot(
  out_ser$time,
  out_ser[, nn <- paste0(sub_ref$subs, '_conc')],
  col = cc <- c('blue', 'orange', 'green', 'black', 'red'),
  lty = 1:length(nn),
  type = 'l',
  ylab = 'Substrate conc.'
)
legend('topright', legend = nn, col = cc, lty = 1:length(nn))
```



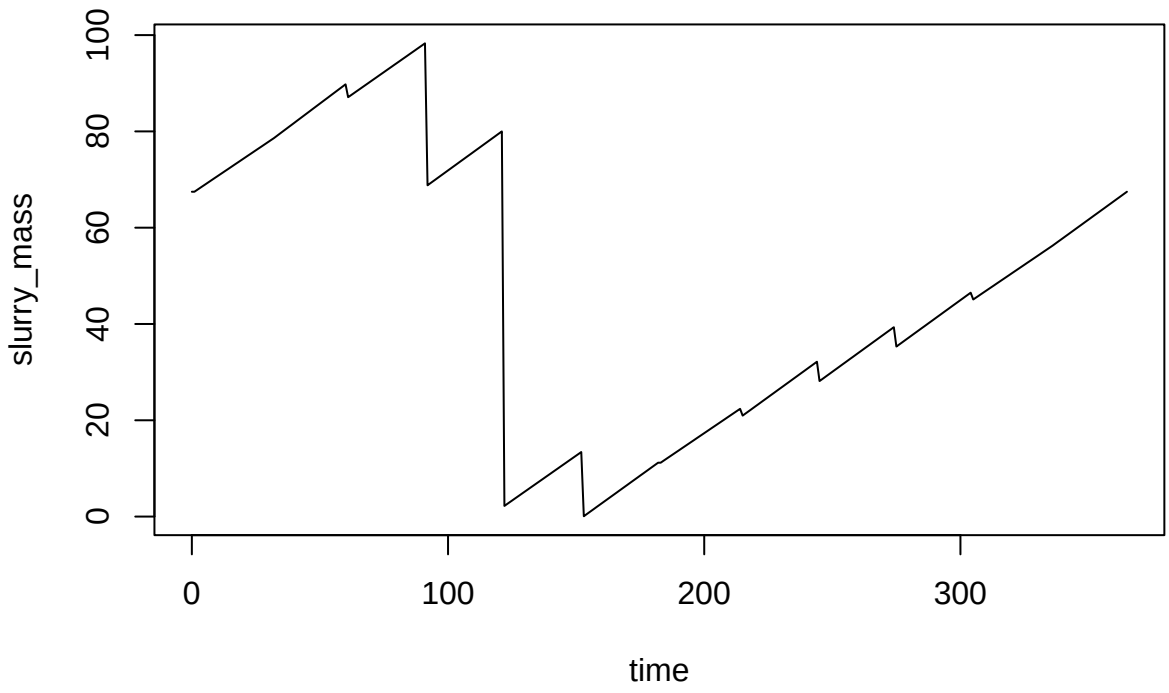

```
plot(VFA ~ time, data = out_ser, type = 'l')
```



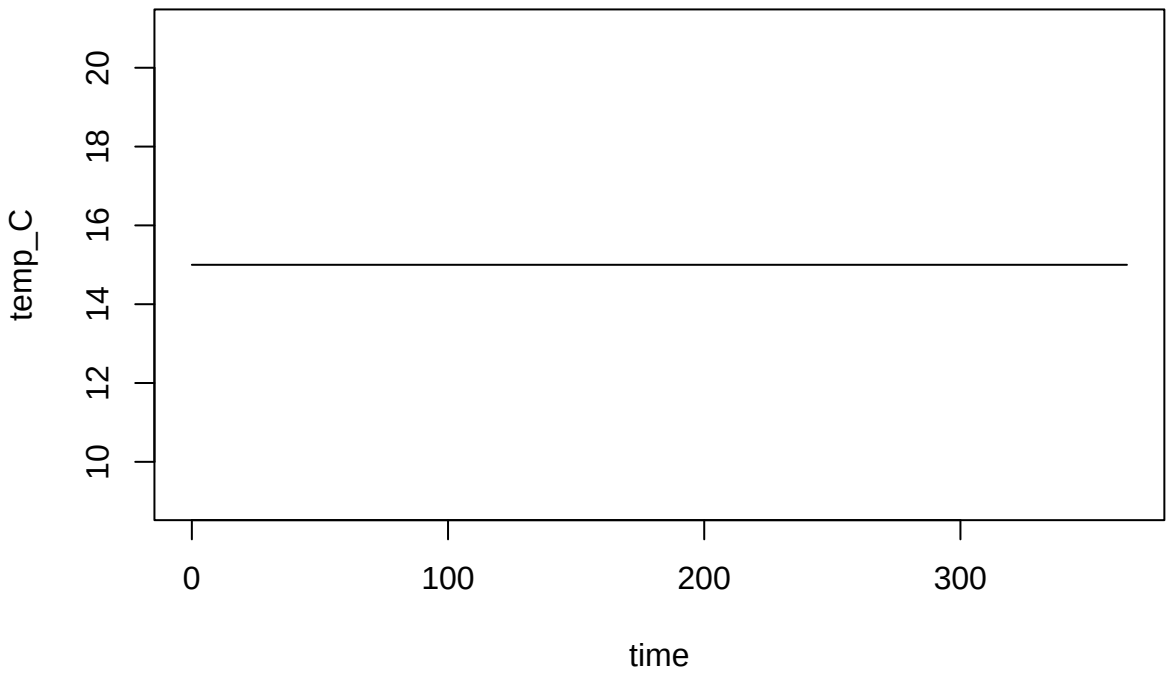

```
plot(VFA_conc ~ time, data = out_ser, type = 'l')
```




```
plot(slurry_mass ~ time, data = out_ser, type = 'l')
```




```
plot(temp_C ~ time, data = out_ser, type = 'l')
```

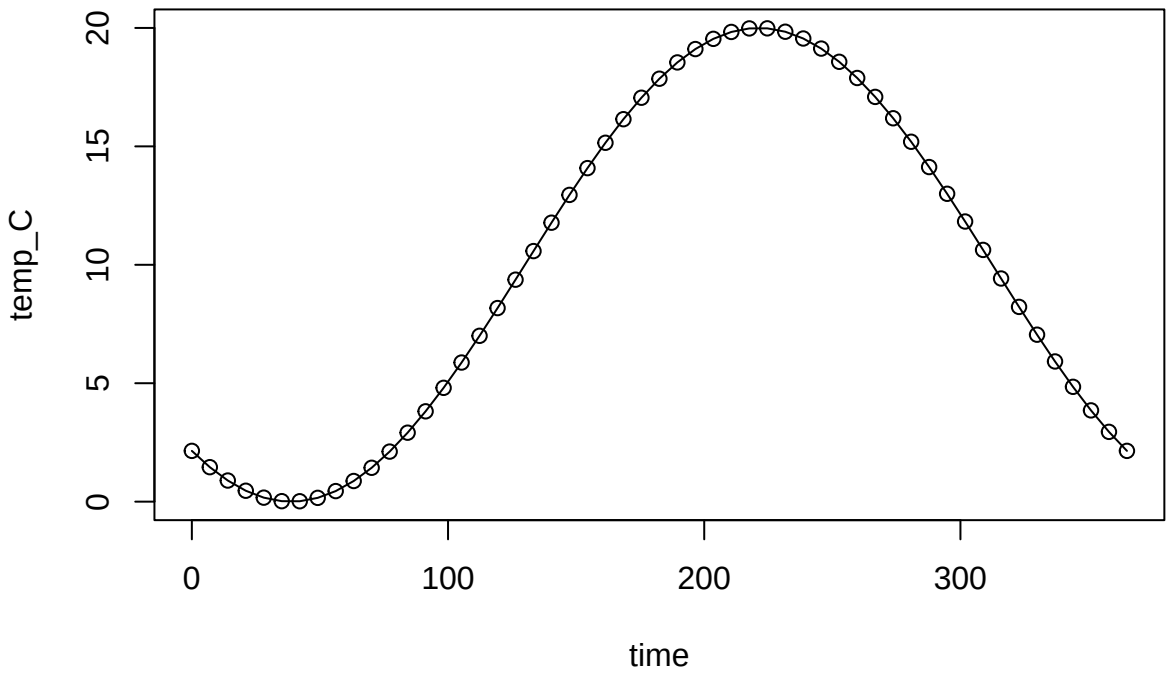


Time-variable inputs

```
temp_dat <- data.frame(time = seq(0, 365, length.out = 53))

temp_dat$temp_C <- 10 + 10 * sin((temp_dat$time - 130) / 365 * 2 * pi)

plot(temp_C ~ time, data = temp_dat, type = 'o')
```




```
devtools::load_all()
```

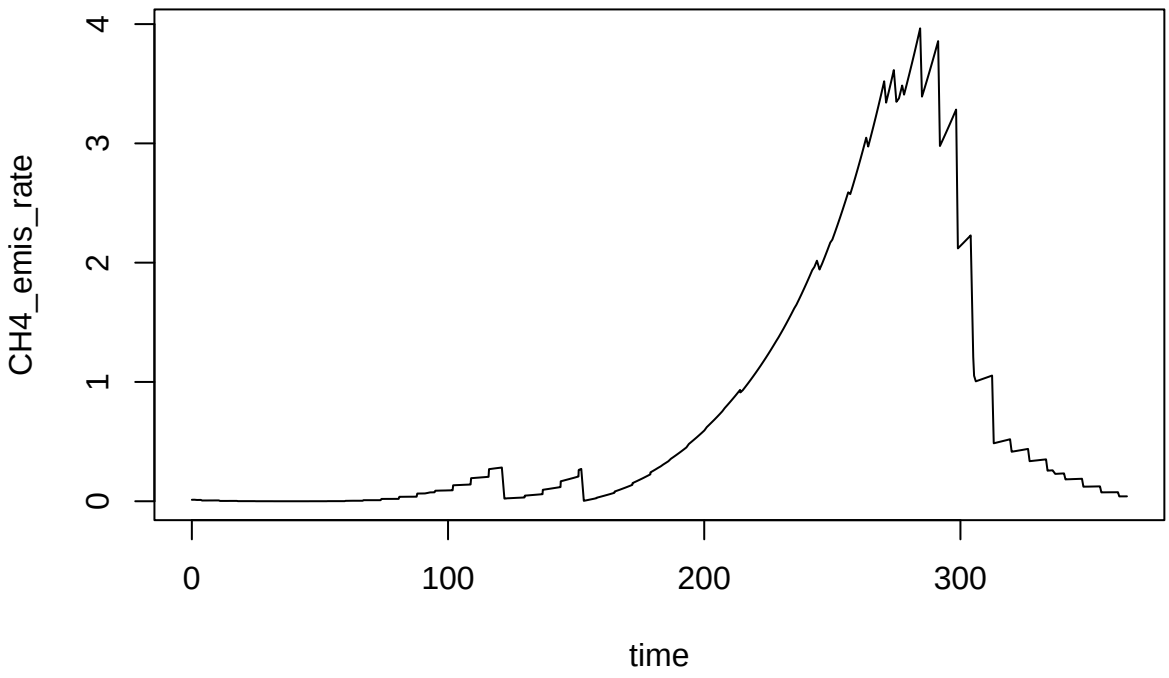
```
## i Loading abmneo
```

```
out_var <- abmneo(
  storage = stor_ser_ref,
  365,
  inf_pars = inf_ref,
  grp_pars = grp_ref,
  sub_pars = sub_ref,
  var_pars = list(var = temp_dat),
)
```

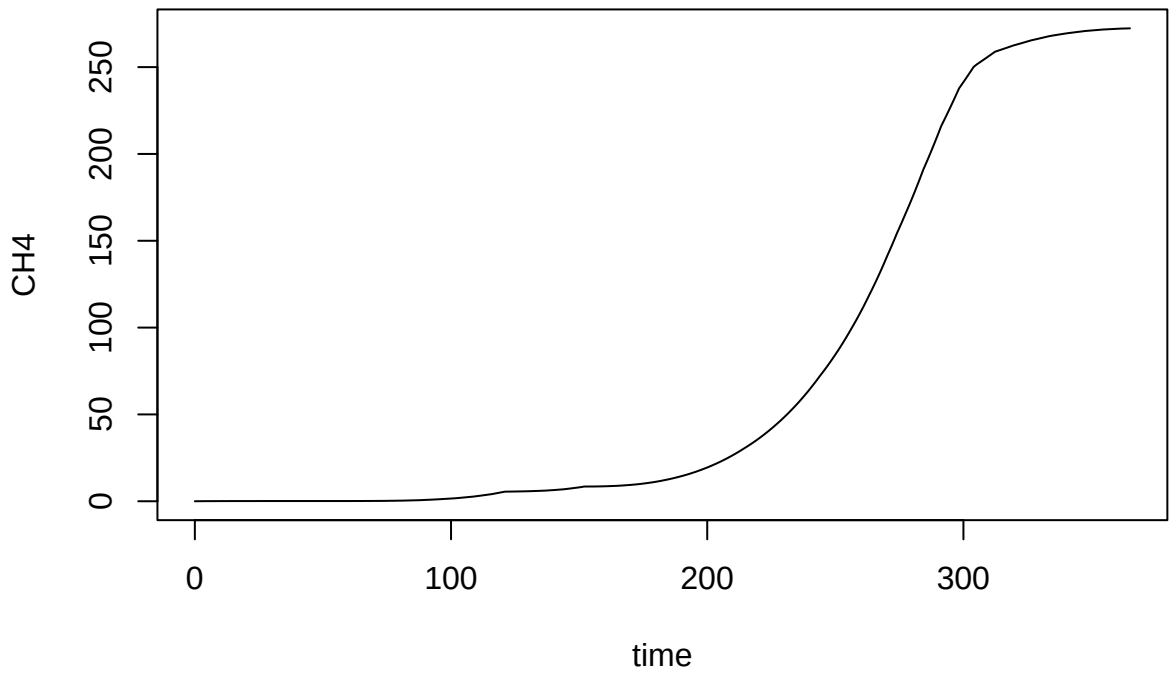
```
## Warning in pack_pars(storage = storage, inf_pars = inf_pars, grp_pars =
## grp_pars, : Expanding mstoich matrix with new rows to include missing comps or
## gases.
```

And the same plots again.

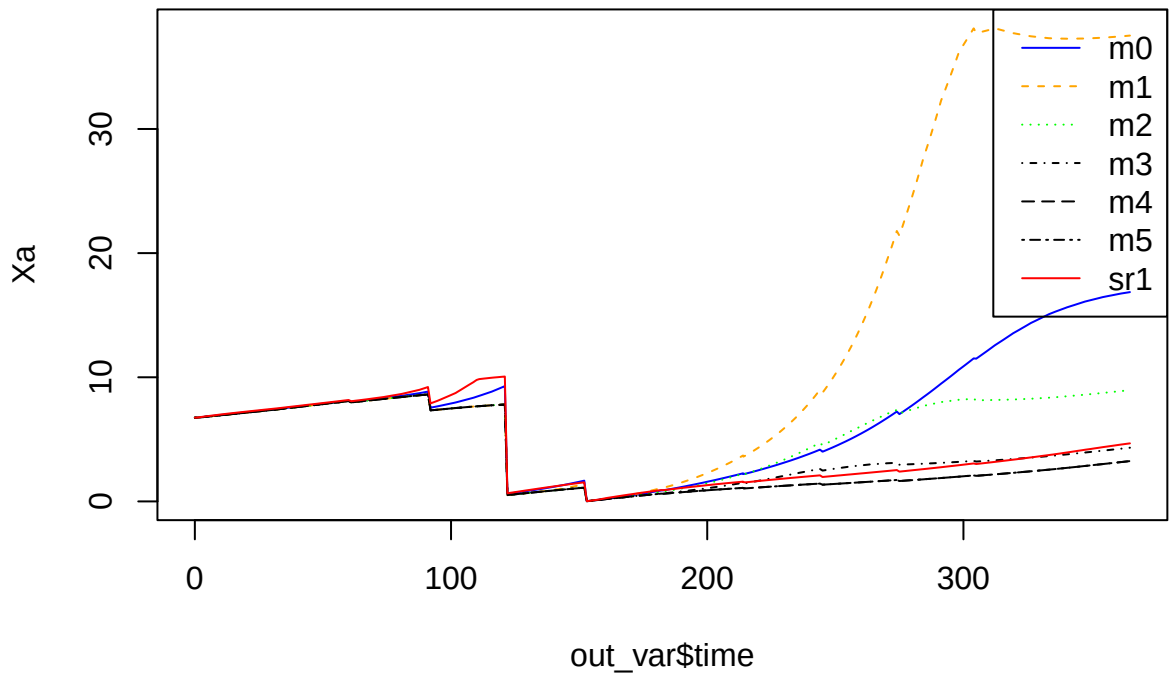
```
plot(CH4_emis_rate ~ time, data = out_var, type = 'l')
```



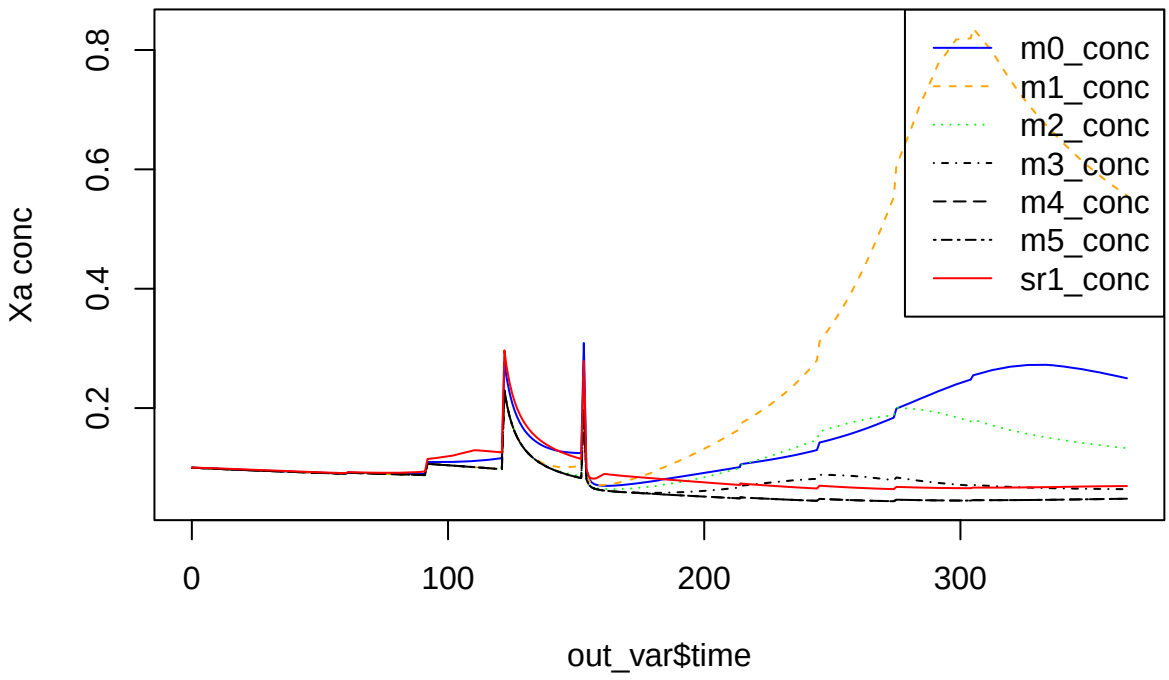

```
plot(CH4 ~ time, data = out_var, type = 'l')
```



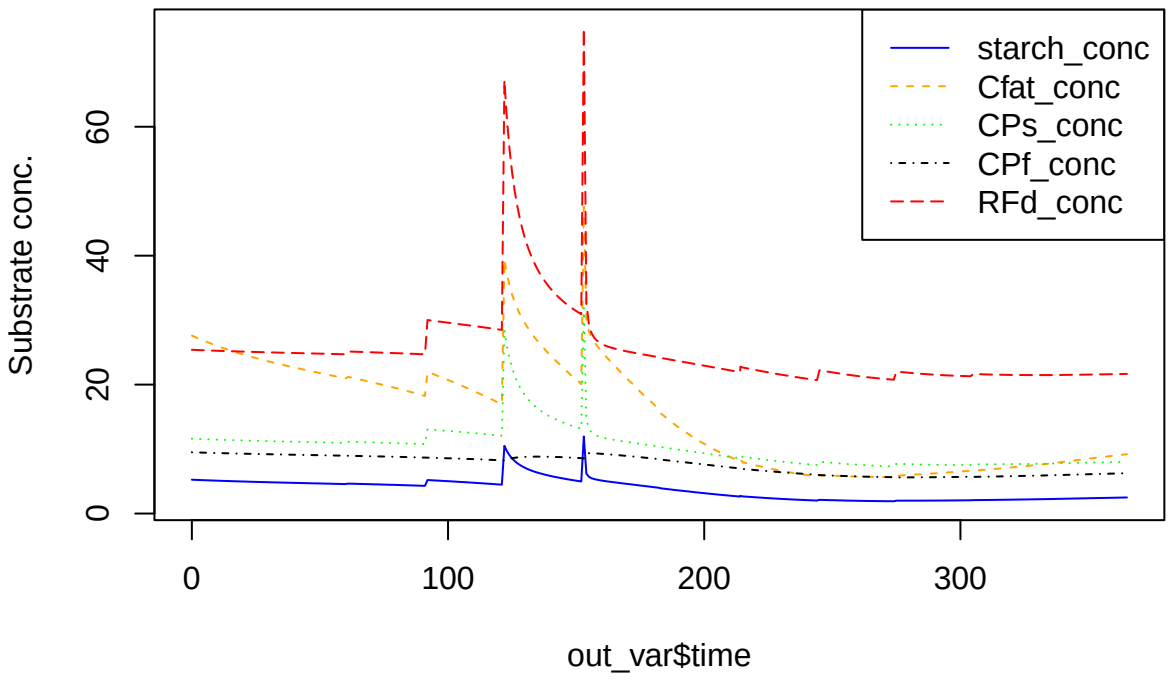

```
matplot(
  out_var$time,
  out_var[, nn <- grp_ref$grps],
  col = cc <- c('blue', 'orange', 'green', 'black', 'black', 'black', 'red'),
  lty = 1:length(nn), type = 'l',
  ylab = 'Xa'
)
legend('topright', legend = nn, col = cc, lty = 1:length(nn))
```



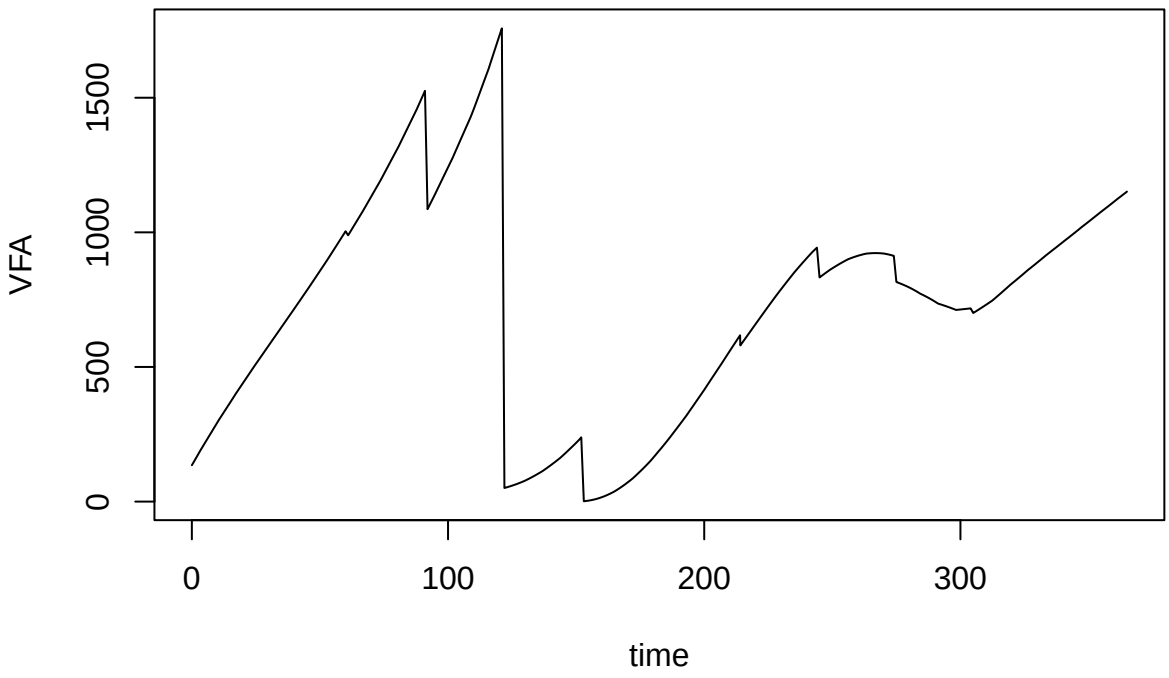

```
matplot(
  out_var$time,
  out_var[, nn <- paste0(grp_ref$grps, '_conc')],
  col = cc <- c('blue', 'orange', 'green', 'black', 'black', 'black', 'red'),
  lty = 1:length(nn),
  type = 'l',
  ylab = 'Xa conc'
)
legend('topright', legend = nn, col = cc, lty = 1:length(nn))
```



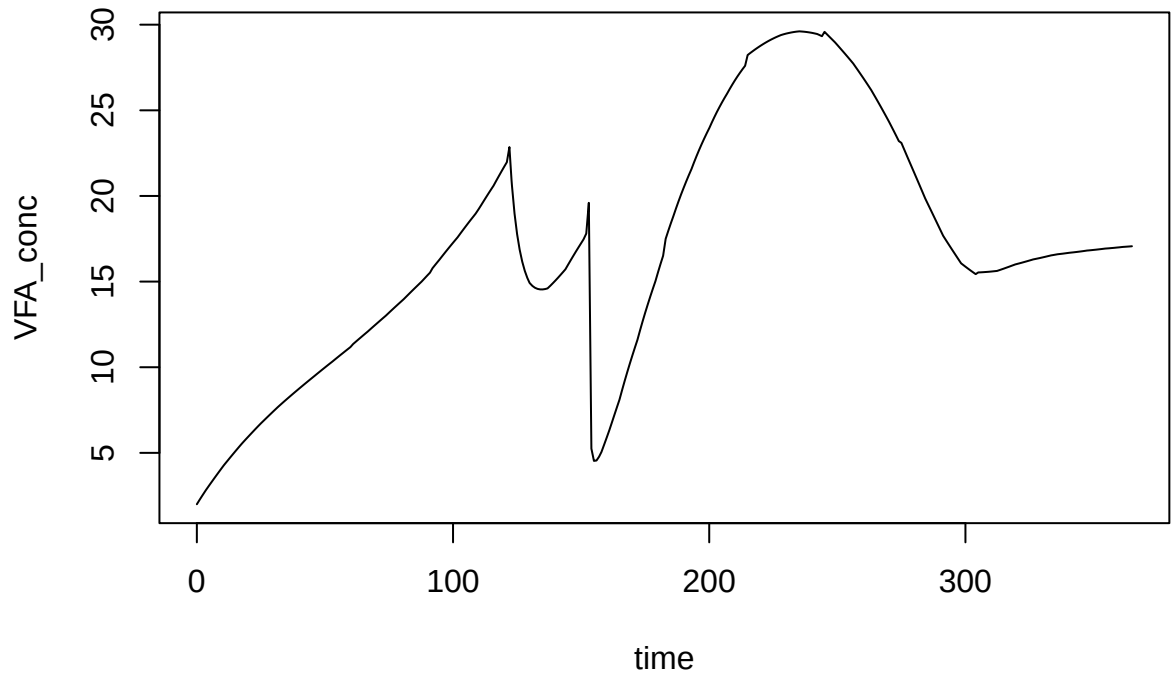

```
matplot(
  out_var$time,
  out_var[, nn <- paste0(sub_ref$subs, '_conc')],
  col = cc <- c('blue', 'orange', 'green', 'black', 'red'),
  lty = 1:length(nn),
  type = 'l',
  ylab = 'Substrate conc.'
)
legend('topright', legend = nn, col = cc, lty = 1:length(nn))
```



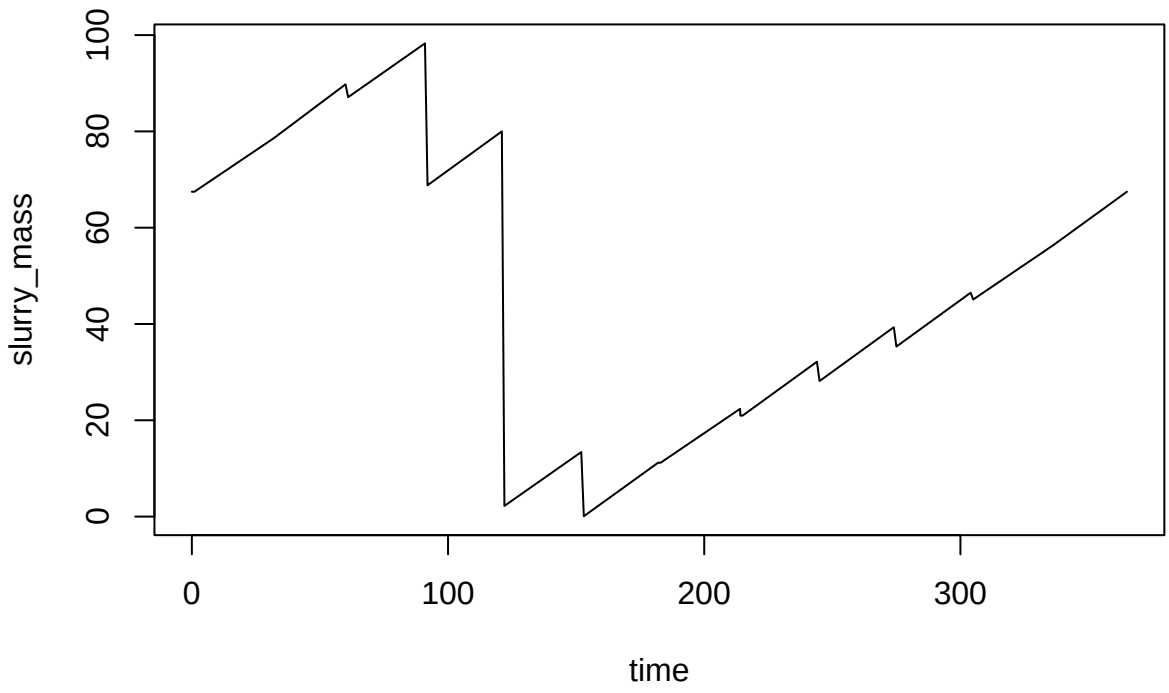

```
plot(VFA ~ time, data = out_var, type = 'l')
```



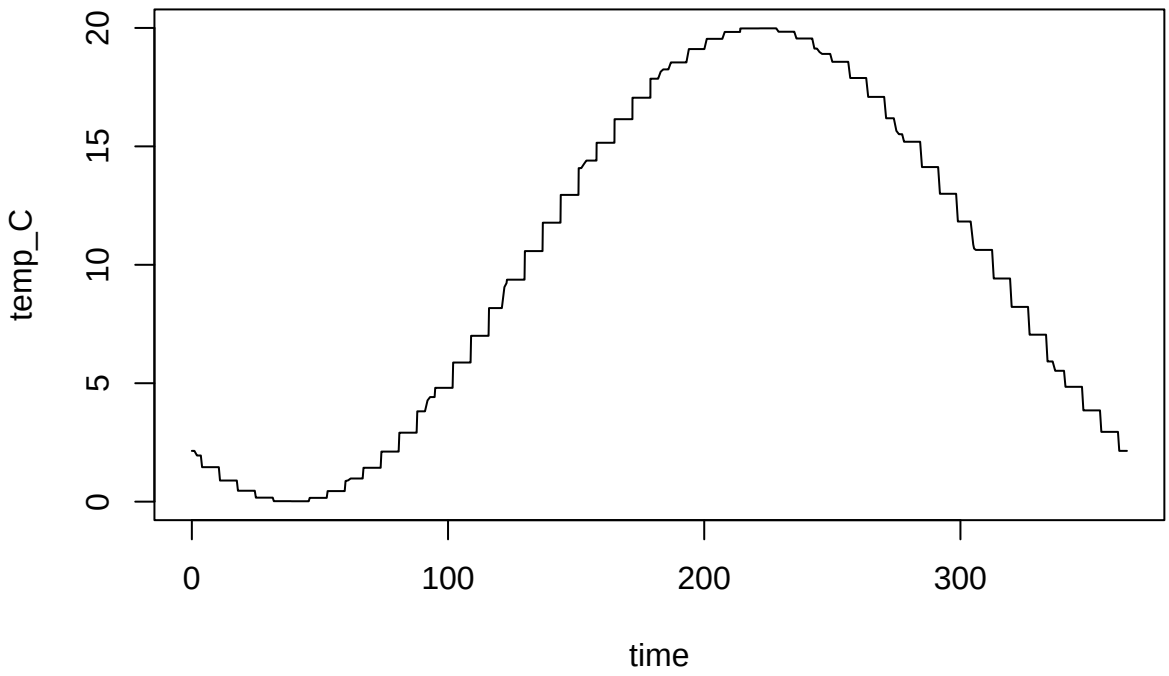

```
plot(VFA_conc ~ time, data = out_var, type = 'l')
```




```
plot(slurry_mass ~ time, data = out_var, type = 'l')
```




```
plot(temp_C ~ time, data = out_var, type = 'l')
```


temp_C
0
5
10
15
20
0
100
200
300
time